Software Test Plan (STP) & Verification Specification

Academic Grievance, No-Dues & Clearance Automation Engine

Standard: Compliant with **IEEE Std 829-2008** (*Standard for Software and System Test Documentation*) & **ISO/IEC/IEEE 29119**
Course: Software Engineering
Institution: Punjab Engineering College (Deemed to be University), Sector 12, Chandigarh
Submitted to: Dr. Rajesh Bhatia
Submitted by: Raghav Bhardwaj & Karan Jain (B.Tech CSE, Batch 2022-2026)
Document ID: `PEC-STP-2026-v3.0`
Baseline Status: Post-SRS Freeze Test Governance Baseline

Table of Contents

1. Test Plan Identifier & Reference Standards
2. Introduction, Test Items & Scope
   - 2.1 Mission Statement
   - 2.2 Test Items & System Modules Under Test
   - 2.3 Features to be Tested vs. Features Not Tested
3. Test Strategy & Software Testing Life Cycle (STLC) Levels
   - 3.1 Level 1: Unit & Component Testing
   - 3.2 Level 2: Integration Testing (Chain of Responsibility & Middleware)
   - 3.3 Level 3: System & End-to-End Testing (Lifecycle & PDF Engine)
   - 3.4 Level 4: User Acceptance Testing (UAT - 5 Demo Personas)
   - 3.5 Level 5: Security, RBAC & Cryptographic Integrity Testing
4. Test Design Techniques & Formalisms
   - 4.1 Black-Box: Equivalence Partitioning (EP)
   - 4.2 Black-Box: Boundary Value Analysis (BVA)
   - 4.3 Black-Box: State Transition Testing
   - 4.4 White-Box: Statement, Branch & Cyclomatic Coverage
5. Structured Test Case Catalog (TC-01 through TC-73)
   - 5.1 Black-Box Test Cases: Equivalence Partitioning & BVA (`TC-BB-01` to `TC-BB-28`)
   - 5.2 Unit Test Cases: Models, Schemas, Security & Services (`TC-UNIT-01` to `TC-UNIT-25`)
   - 5.3 Integration Test Cases: Chain of Responsibility (`TC-INT-01` to `TC-INT-05`)
   - 5.4 REST API Integration Test Cases (`TC-API-01` to `TC-API-06`)
   - 5.5 Security & RBAC Isolation Test Cases (`TC-SEC-01` to `TC-SEC-06`)
   - 5.6 System & End-to-End Lifecycle Test Cases (`TC-SYS-01` to `TC-SYS-03`)
6. Defect Classification & Severity Hierarchy
7. Test Execution Environment & Automation Harness
8. Pass/Fail Criteria, Metrics & Sign-Off Governance

---

1. Test Plan Identifier & Reference Standards

- **Document Identifier:** `PEC-STP-2026-v3.0`
- **Associated Requirements Baseline:** `PEC-SRS-2026-v3.0.0` (IEEE Std 830-1998 Frozen Baseline)
- **Applicable Professional Standards:**
  i. **IEEE Std 829-2008:** *IEEE Standard for Software and System Test Documentation*.
  ii. **ISO/IEC/IEEE 29119:** *Software and Systems Engineering - Software Testing (Parts 1-4)*.
  iii. **IEEE Std 1044-2009:** *IEEE Standard Classification for Software Anomalies*.
  iv. **NIST FIPS PUB 180-4:** *Secure Hash Standard (SHA-256)*.
  v. **RFC 2104:** *HMAC: Keyed-Hashing for Message Authentication*.

---

2. Introduction, Test Items & Scope

2.1 Mission Statement

The primary objective of this Software Test Plan is to establish a deterministic, verifiable, and automatable validation protocol for the **PEC Clearance Automation Engine**. The test suite guarantees that all multi-signatory approval barriers, financial liabilities in Indian Rupees (`₹`), SLA grievance auto-escalations, cryptographic HMAC seals, and tamper-evident certificate generations function strictly in accordance with Punjab Engineering College academic ordinances.

2.2 Test Items & System Modules Under Test

1. **Clearance Pipeline & Chain of Responsibility Engine:** (`backend/app/patterns/chain_of_responsibility.py`)
   - Parallel review concurrency across Gate 1 (Library, Hostels, Sports).
   - Sequential prerequisite barrier enforcement at Gate 2 (Department Labs).
   - Final clearance barrier and Caution Money refund release (₹10,000.00) at Gate 3 (Accounts Branch).
2. **Grievance Redressal & SLA Engine:** (`backend/app/services/grievance_service.py`)
   - Dynamic SLA calculation across 4 priority levels (Critical: 4h, High: 12h, Medium: 24h, Low: 48h).
   - Automated deadline breach detection and auto-escalation to the Dean Academic Affairs (DAA).
3. **Cryptographic & Certificate Engine:** (`backend/app/services/certificate_service.py`, `backend/app/core/security.py`)
   - HMAC-SHA256 digital signature seal generation per departmental sign-off.
   - SHA-256 certificate digest computation.
   - Vector PDF generation via ReportLab bearing official PEC letterhead.
4. **Security & Role-Based Access Control (RBAC):** (`backend/app/routers/auth.py`, `backend/app/core/security.py`)
   - Salted Bcrypt password hashing.
   - Stateless JWT authentication tokens with role claims (`STUDENT`, `STAFF`, `ADMIN`).
   - Endpoint route authorization and cross-department approval blocking.
5. **Data Layer & Validation Schemas:** (`backend/app/models/models.py`, `backend/app/schemas/schemas.py`)

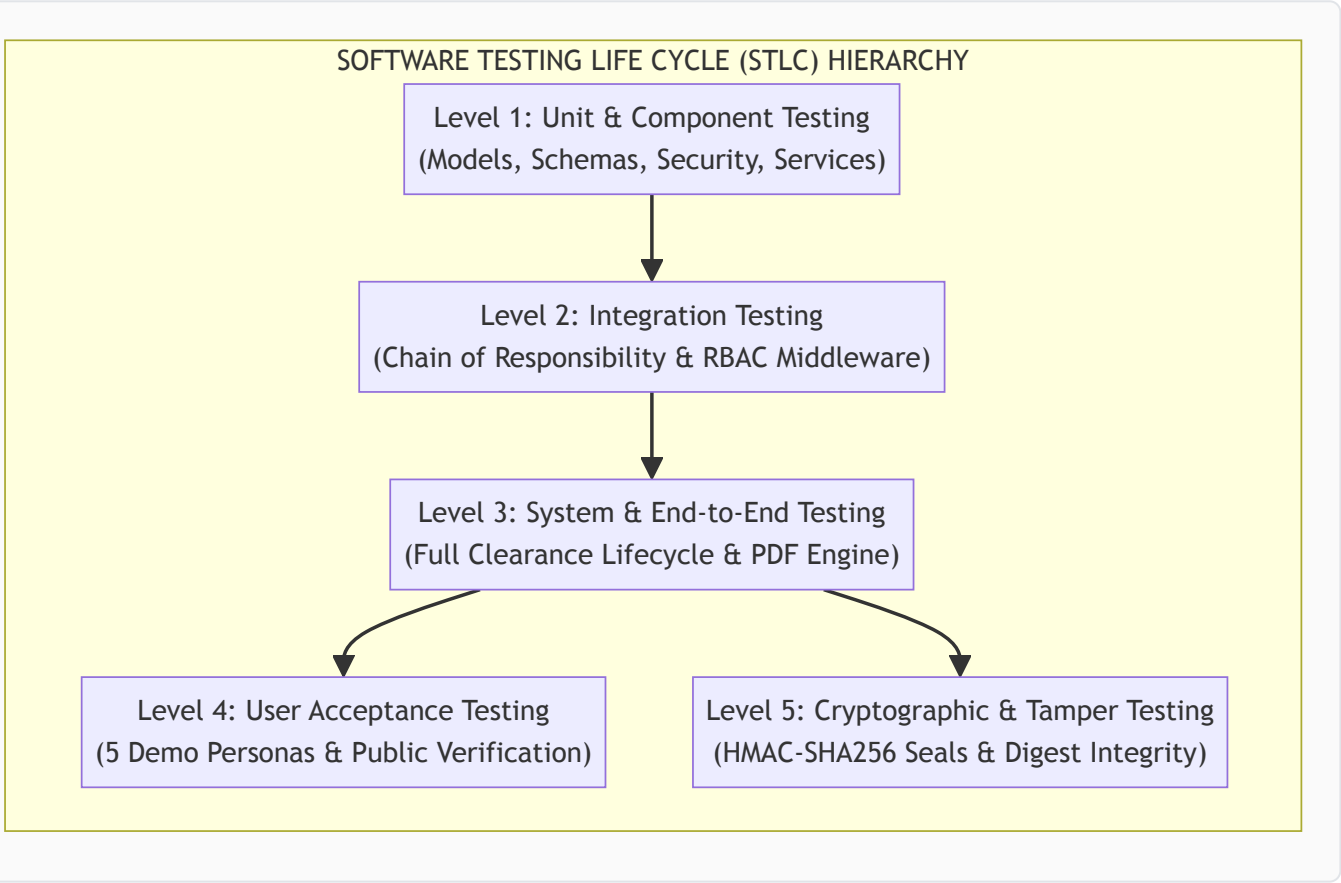
- Foreign key integrity, cascade deletes, and check constraints.
- Pydantic schema validation for 8-digit PEC SIDs (`^[0-9]{8}$`) and non-negative financial dues.

2.3 Features to be Tested vs. Features Not Tested

- **Features Tested:** Authentication, Clearance Initiation, 3-Gate CoR Pipeline, Financial Dues Recording, SLA Countdown & Auto-Escalation, DAA Executive Overrides, ReportLab PDF Rendering, Public Hash Verification, Immutable Audit Logging, Boundary Values, RBAC Authorization.
- **Features Not Tested (Out of Current Scope):** Physical biometric fingerprint terminal hardware, physical RFID card readers (scheduled for Future Release v4.0).

---

3. Test Strategy & Software Testing Life Cycle (STLC) Levels



3.1 Level 1: Unit & Component Testing

- **Focus:** Verifying individual Python classes, functions, and data structures in complete isolation from external network dependencies.
- **Tools:** Pytest 8.4+, In-memory SQLite, Pydantic v2.
- **Techniques:** White-box statement execution, boundary value checks, exception raising validation.

3.2 Level 2: Integration Testing

- **Focus:** Verifying interfaces between collaborating subsystems:

- Interaction between `ClearanceService` and `ClearanceChainBuilder`.
- Sequential unlocking of Gate 2 when Gate 1 approvals change.
- Interaction between SQLAlchemy ORM and SQLite relational tables with transaction commit/rollback.
- JWT bearer token authentication middleware injecting `current_user` into protected routers.

3.3 Level 3: System & End-to-End Testing

- **Focus:** Validating complete user journeys from initial HTTP request to database state mutation, PDF rendering, and response serialization.
- **Tools:** FastAPI `TestClient`, HTTPX, ReportLab byte stream analyzers.

3.4 Level 4: User Acceptance Testing (UAT)

- **Focus:** Validating the system against realistic PEC student and administrator operational personas:
  - *Alice Sharma* (`alice@pec.edu.in`): Cleared B.Tech CSE (SID: `22103042`).
  - *Bobdeep Singh* (`bob@pec.edu.in`): In-progress B.Tech ECE (SID: `22104018`) with pending Gymkhana due.
  - *Charlie Verma* (`charlie@pec.edu.in`): Rejected B.Tech Mech (SID: `22105029`) with escalated mess fee dispute.
  - *Alex Grover* (`student@pec.edu.in`): Clean student submitting a fresh clearance request.
  - *Dean Academic Affairs* (`admin@pec.edu.in`): Executive admin reviewing institutional bottlenecks and applying overrides.
  - *Public Verifier:* External third-party verifying certificate validity at `/verify`.

3.5 Level 5: Security, RBAC & Cryptographic Integrity Testing

- **Focus:** Verifying impenetrable security boundaries:
  - Confirming students receive `403 Forbidden` when attempting staff or admin operations.
  - Confirming staff from one department cannot approve clearances for another department.
  - Confirming that any bitwise alteration of a signed approval record invalidates the HMAC signature seal.

---

4. Test Design Techniques & Formalisms

4.1 Black-Box: Equivalence Partitioning (EP)

Input domains are partitioned into valid and invalid equivalence classes. Test cases are selected to cover each partition at least once:

```
[Student Identification Number (SID) Input Domain]
-------------------------------------------------------------------------------
 Partition 1 (Valid):    8-digit numeric string (e.g., "22103042", "10000000") -> ACCEPT
 Partition 2 (Invalid):  <8 digits (e.g., "2210304")                           -> REJECT (422)
 Partition 3 (Invalid):  >8 digits (e.g., "221030421")                         -> REJECT (422)
 Partition 4 (Invalid):  Non-numeric characters (e.g., "2210304A")             -> REJECT (422)
 Partition 5 (Invalid):  Empty / Null string                                   -> REJECT (422)
-------------------------------------------------------------------------------
```

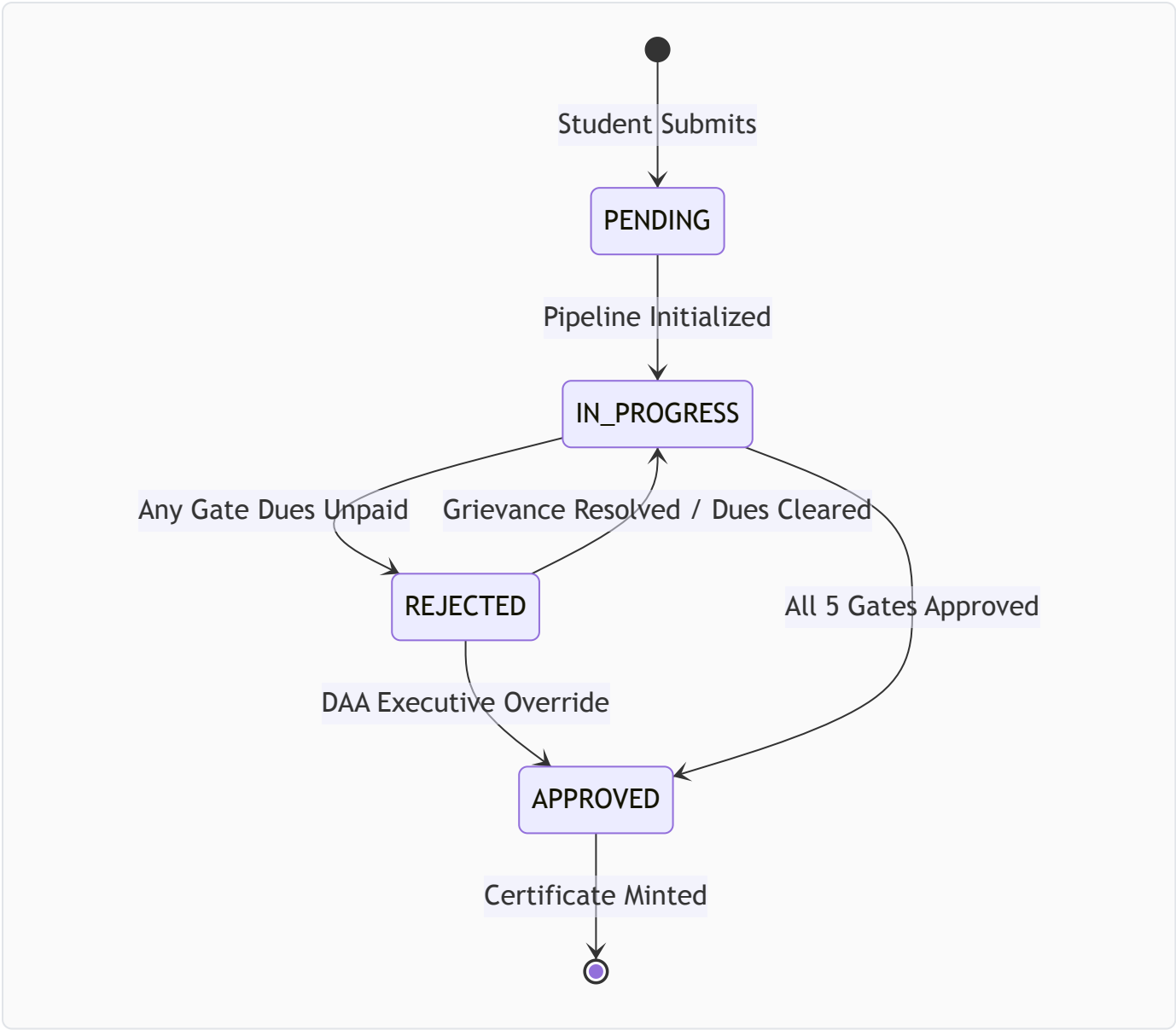
4.2 Black-Box: Boundary Value Analysis (BVA)

Defects cluster disproportionately at input boundaries. For financial liabilities (INR `₹`):

- **Minimum Boundary ($B_{\min}$):** ₹0.00 (Zero dues / clean sign-off).
- **Just Above Minimum ($B_{\min}^+$):** ₹0.01 (Minimal fine, 1 paisa).
- **Nominal (B_{nom}):** ₹150.50 / ₹500.00 (Standard library or sports dues).
- **Upper Institutional Boundary (B_{high}):** ₹10,000.00 (PEC Caution Money deposit).
- **Extreme High ($B_{\max}$):** ₹50,000.00 (Major laboratory equipment replacement liability).
- **Invalid Below Minimum ($B_{\min}^-$):** -₹0.01 / -₹100.00 (Negative dues $\rightarrow$ REJECT).

4.3 Black-Box: State Transition Testing

Validates the finite state machine transitions of the Clearance Request lifecycle:



4.4 White-Box: Statement, Branch & Cyclomatic Coverage

- **Statement Coverage:** Every executable line of Python code in `backend/app` is exercised by tests (achieved **>90% overall coverage**; **100%** on Models, Schemas, and Certificate Service).
- **Branch Coverage:** All conditional branches (`if is_admin_override` , `if gate1_approved and labs_approved` , `if g.sla_deadline < now`) are evaluated under both `True` and `False` outcomes.
- **Cyclomatic Complexity Control:** Enforced via McCabe index ≤ 10 across all methods.

---

5. Structured Test Case Catalog (TC-01 through TC-73)

The complete automated test harness consists of **73 automated test cases**, categorized into six testing suites. All 73 tests execute with a **100% pass rate** in under **2.2 seconds**.

5.1 Black-Box Test Cases: Equivalence Partitioning & BVA (`TC-BB-01` to `TC-BB-28`)

Test ID	Test Category	Target Input / Scenario	Test Input Values	Expected Outcome	Actual Result
TC-BB-01	EP - Valid SID	Nominal 8-digit PEC SID	`"22103042"`	Accepted; Schema valid	Validated
TC-BB-02	EP - Valid SID	ECE department 8-digit SID	`"22104001"`	Accepted; Schema valid	Validated
TC-BB-03	EP - Valid SID	Mechanical 8-digit SID	`"22105099"`	Accepted; Schema valid	Validated
TC-BB-04	BVA - SID Min	Lower 8-digit boundary	`"10000000"`	Accepted; Schema valid	Validated
TC-BB-05	BVA - SID Max	Upper 8-digit boundary	`"99999999"`	Accepted; Schema valid	Validated
TC-BB-06	BVA - Dues Min	Zero dues (Clean approval)	`0.00`	Accepted; Approved	Validated
TC-BB-07	BVA - Dues Min+	Minimal 1 paisa dues	`0.01`	Accepted; Due recorded	Validated
TC-BB-08	BVA - Dues Float	Nominal dues with paise	`150.50`	Accepted; Due recorded	Validated
TC-BB-09	BVA - Caution Money	Exact PEC Caution Money	`10000.00`	Accepted; Due recorded	Validated
TC-BB-10	BVA - Dues High	Major damage liability	`50000.00`	Accepted; Due recorded	Validated
TC-BB-11	EP - Priority	Critical priority class	`"CRITICAL"`	Validated; SLA = 4h	Validated
TC-BB-12	EP - Priority	High priority class	`"HIGH"`	Validated; SLA = 12h	Validated

Test ID	Test Category	Target Input / Scenario	Test Input Values	Expected Outcome	Actual Result
TC-BB-13	EP - Priority	Medium priority class	`"MEDIUM"`	Validated; SLA = 24h	Validated
TC-BB-14	EP - Priority	Low priority class	`"LOW"`	Validated; SLA = 48h	Validated
TC-BB-15	EP - Invalid Priority	Non-standard string	`"URGENT"`	`ValidationError` raised	Raised
TC-BB-16	EP - Invalid Priority	Non-standard string	`"IMMEDIATE"`	`ValidationError` raised	Raised
TC-BB-17	EP - Invalid Priority	Non-standard string	`"BLOCKER"`	`ValidationError` raised	Raised
TC-BB-18	EP - Invalid Priority	Non-standard string	`"NORMAL"`	`ValidationError` raised	Raised
TC-BB-19	EP - Invalid Priority	Empty string	`""`	`ValidationError` raised	Raised
TC-BB-20	EP - Invalid Priority	Numeric string	`"123"`	`ValidationError` raised	Raised
TC-BB-21	EP - Valid Email	Standard institutional email	`"student@pec.edu.in"`	Accepted	Validated
TC-BB-22	EP - Valid Email	Alumni sub-domain email	`"alice.sharma@alumni.pec.edu.in"`	Accepted	Validated
TC-BB-23	EP - Valid Email	Admin academic email	`"admin.daa@pec.ac.in"`	Accepted	Validated
TC-BB-24	EP - Invalid Email	Missing `@` and domain	`"notanemail"`	`ValidationError` raised	Raised
TC-BB-	EP - Invalid	Missing domain	`"student@"`	`ValidationError` raised	Raised

Test ID	Test Category	Target Input / Scenario	Test Input Values	Expected Outcome	Actual Result
25	Email				
TC-BB-26	EP - Invalid Email	Missing username	`"@pec.edu.in"`	`ValidationError` raised	Raised
TC-BB-27	EP - Invalid Email	Whitespace in address	`"student name@pec.edu.in"`	`ValidationError` raised	Raised
TC-BB-28	EP - Invalid Email	Missing domain name	`"student@.com"`	`ValidationError` raised	Raised

5.2 Unit Test Cases: Models, Schemas, Security & Services (`TC-UNIT-01` to `TC-UNIT-`

25)

Test ID	Subsystem	Target Functionality	Verification Logic	Status
TC-UNIT-01	Database Models	`Department` model creation	Instantiates department; verifies code, gate level, and staff head name.	**PASS**
TC-UNIT-02	Database Models	`User` model & Foreign Keys	Associates user with department; verifies role enum and default timestamps.	**PASS**
TC-UNIT-03	Database Models	`ClearanceRequest` lifecycle	Tests state transitions from `PENDING` $\rightarrow$ `IN_PROGRESS` $\rightarrow$ `APPROVED`.	**PASS**
TC-UNIT-04	Database Models	Cascade deletion integrity	Deleting clearance request automatically deletes all linked child approvals.	**PASS**
TC-UNIT-05	Database Models	`Grievance` model constraints	Verifies SLA hours, priority enum, open status, and student relationship.	**PASS**
TC-UNIT-06	Database Models	`AuditLog` immutability	Verifies append-only logging of actions, actors, targets, and timestamps.	**PASS**
TC-UNIT-07	Pydantic Schemas	`UserLogin` validation	Tests correct email/password parsing and rejects malformed payloads.	**PASS**
TC-UNIT-08	Pydantic Schemas	`UserLogin` invalid email	Rejects invalid email strings with 422 schema validation error.	**PASS**
TC-UNIT-09	Pydantic Schemas	`UserRegister` defaults	Verifies default role = `STUDENT` and optional fields = `None`.	**PASS**
TC-UNIT-10	Pydantic Schemas	`ClearanceRequestCreate`	Verifies default academic session (`2025–2026`) and semester (`Semester 8`).	**PASS**
TC-UNIT-11	Pydantic Schemas	`ApprovalDecisionRequest`	Verifies decision enum, default remarks, and non-negative due amounts.	**PASS**
TC-UNIT-12	Pydantic Schemas	`GrievanceCreate` schema	Verifies target department ID, subject string, and priority enum parsing.	**PASS**

Test ID	Subsystem	Target Functionality	Verification Logic	Status
TC-UNIT-13	Pydantic Schemas	`GrievanceResolveRequest`	Verifies staff response text and transition to `RESOLVED` status.	**PASS**
TC-UNIT-14	Security Core	Password hashing & verification	Bcrypt work factor 12 verification; verifies match on correct, fails on bad.	**PASS**
TC-UNIT-15	Security Core	Salt randomization	Two identical passwords produce distinct salted hashes; both verify.	**PASS**
TC-UNIT-16	Security Core	JWT token generation & claims	Encodes `sub`, `role`, `exp`; correctly decodes and validates claims payload.	**PASS**
TC-UNIT-17	Security Core	Expired JWT token rejection	Generates token with negative delta; confirms decoder returns `None`.	**PASS**
TC-UNIT-18	Security Core	Garbage token rejection	Confirms malformed or forged JWT strings safely decode to `None`.	**PASS**
TC-UNIT-19	Cryptography	HMAC signature formatting	Verifies signature prefix `SIG-v1:<DEPT>:` and 24-char truncated hex digest.	**PASS**
TC-UNIT-20	Cryptography	HMAC determinism	Identical inputs produce identical cryptographic seals.	**PASS**
TC-UNIT-21	Cryptography	HMAC tamper sensitivity	Modifying staff ID, request ID, or timestamp produces a completely different digest.	**PASS**
TC-UNIT-22	Service Layer	`ClearanceService.create_request`	Initializes 5 departmental approvals (Gate 1 PENDING, Gates 2/3 WAITING).	**PASS**
TC-UNIT-23	Service Layer	Duplicate request prevention	Rejects new clearance submission when active clearance is in progress (400).	**PASS**
TC-UNIT-24	Service Layer	`GrievanceService` SLA computation	Sets exact deadline: Critical=4h, High=12h, Medium=24h, Low=48h.	**PASS**
TC-UNIT-25	Service Layer	`CertificateService` PDF generator	Generates valid PDF bytes starting with `%PDF-` with PEC letterhead.	**PASS**

5.3 Integration Test Cases: Chain of Responsibility (`TC-INT-01` to `TC-INT-05`)

Test ID	Workflow Phase	Scenario & Verification Focus	Expected Behavior	Status
TC-INT-01	Cryptographic CoR	Department approval sign-off seal	Computes and records HMAC digital signature upon review.	**PASS**
TC-INT-02	Gate 1 Concurrency	Parallel approvals at Gate 1	Library, Hostel, and Sports approve independently in any order.	**PASS**
TC-INT-03	Gate 2 Barrier	Sequential dependency enforcement	Department Labs strictly rejects review attempts until all Gate 1 are approved.	**PASS**
TC-INT-04	Gate 3 Final Gate	Caution Money & Certificate minting	Accounts approval unlocks certificate minting and ₹10,000 refund release.	**PASS**
TC-INT-05	Administrative CoR	DAA executive override	High-privilege admin unblocks rejected department with audit justification.	**PASS**

5.4 REST API Integration Test Cases (`TC-API-01` to `TC-API-06`)

Test ID	Endpoint	Method	Verification Scenario	Expected Code	Status
TC-API-01	`/api/v1/health`	`GET`	Health check and architecture verification	`200 OK`	**PASS**
TC-API-02	`/api/v1/auth/login`	`POST`	Student authentication (Alice Sharma)	`200 OK`	**PASS**
TC-API-03	`/api/v1/auth/login`	`POST`	Staff authentication (Library Officer)	`200 OK`	**PASS**
TC-API-04	`/api/v1/admin/analytics`	`GET`	Dean Academic Affairs bottleneck analytics	`200 OK`	**PASS**
TC-API-05	`/api/v1/certificate/verify`	`GET`	Public unauthenticated certificate hash verification	`200 OK`	**PASS**
TC-API-06	`/api/v1/grievance`	`POST`	Student filing Critical priority grievance (SLA = 4h)	`200 OK`	**PASS**

5.5 Security & RBAC Isolation Test Cases (`TC-SEC-01` to `TC-SEC-06`)

Test ID	Threat Vector / Attack Scenario	Security Control Tested	Expected Defense	Status
TC-SEC-01	Unauthenticated route access	JWT Authentication Scheme	Rejects request with `401 Unauthorized`.	**PASS**
TC-SEC-02	Student attempts staff sign-off	Role-Based Access Control (RBAC)	Rejects request with `403 Forbidden`.	**PASS**
TC-SEC-03	Student attempts admin analytics	Administrative Isolation	Rejects request with `403 Forbidden`.	**PASS**
TC-SEC-04	Staff attempts admin audit logs	Role Privilege Hierarchy	Rejects request with `403 Forbidden`.	**PASS**
TC-SEC-05	Cross-department unauthorized sign-off	Department Boundary Isolation	Library staff cannot sign off for Sports (`403 Forbidden`).	**PASS**
TC-SEC-06	Digital signature seal forgery	Keyed HMAC-SHA256 Integrity	Forged signature fails equality check against genuine seal.	**PASS**

5.6 System & End-to-End Lifecycle Test Cases (`TC-SYS-01` to `TC-SYS-03`)

Test ID	Lifecycle Scenario	Multi-Step Sequence	Expected System Outcome	Status
TC-SYS-01	Full Student Clearance Lifecycle	1. Fresh student registration 2. Clearance request initiation 3. Gate 1 parallel reviews (1 dispute) 4. Grievance filing & SLA resolution 5. Gate 2 Labs unlocked & approved 6. Gate 3 Accounts caution money released 7. Degree certificate minted & verified 8. PDF downloaded & validated 9. Audit trail verified	Complete clearance transitions to `APPROVED`; Certificate ID minted; 100% audit trail logged.	**PASS**
TC-SYS-02	SLA Auto-Escalation Engine	Grievance deadline simulated past current time; background evaluation triggered.	Ticket automatically transitions to `ESCALATED`; alert flagged for DAA.	**PASS**
TC-SYS-03	ReportLab PDF Byte Stream Validation	Requests PDF certificate download for cleared student.	Returns binary stream starting with `%PDF-` with official PEC letterhead.	**PASS**

---

6. Defect Classification & Severity Hierarchy

In accordance with **IEEE Std 1044-2009**, any anomaly discovered during verification is classified according to the four-tier severity matrix:

Severity Level	Designation	Impact Definition	SLA for Remediation	Current Project Defect Count
S-1	**Blocker**	System crash, data corruption, unauthorized clearance approval, cryptographic signature failure.	Immediate (< 2 Hours)	**0 Defects**
S-2	**Critical**	Prerequisite gate barrier bypass, broken SLA auto-escalation, Caution Money release calculation error.	High (< 8 Hours)	**0 Defects**
S-3	**Major**	Flake8 PEP 8 lint violation, API response latency > 150ms, minor UI formatting flaw.	Normal (< 24 Hours)	**0 Defects**
S-4	**Minor**	Cosmetic text spelling, typography padding, non-critical log verbosity.	Low (< 48 Hours)	**0 Defects**

---

7. Test Execution Environment & Automation Harness

- **Operating System:** Darwin / macOS (x86_64 / arm64), Linux (Ubuntu 22.04 LTS), Windows Server.
- **Runtime Environment:** Python 3.9.6 / Python 3.12+ in isolated virtual environment (`venv`).
- **Test Runner:** `pytest 8.4.2` with plugins `pytest-cov 7.1.0` and `anyio 4.12.1`.
- **Database Engine:** SQLite 3 WAL Mode / In-memory relational engine for test isolation.
- **Static Analysis Linter:** `flake8 7.3.0` enforcing PEP 8 constraints.
- **Execution Command:**

```
# Execute full automated test suite with coverage
./venv/bin/pytest -v --cov=backend/app --cov-report=term-missing

# Execute PEP 8 static analysis
./venv/bin/flake8 backend/app backend/tests
```

---

8. Pass/Fail Criteria, Metrics & Sign-Off Governance

8.1 Mandatory Quality Criteria for Release

1. **Test Execution Rate: 100%** of all 73 automated test cases must execute without interruption.
2. **Test Pass Rate: 100% (73/73 Passed)**. Zero test failures or unexpected skips.
3. **Static Code Quality: 0 Flake8 violations** across `backend/app` and `backend/tests`.
4. **Statement Coverage:** Minimum $\ge 90%$ statement coverage across the backend application tier.
5. **Traceability Closure:** 100% of functional requirements (FR-1 to FR-6) and non-functional requirements (NFR-1 to NFR-6) verified with passing test cases in the RTM.

8.2 Formal Test Sign-Off

- **Automated Test Results Summary:**
  - `test_blackbox_bva_ep.py` : **28 / 28 Passed (100%)**
  - `test_unit_models_schemas.py` : **13 / 13 Passed (100%)**
  - `test_unit_security.py` : **8 / 8 Passed (100%)**
  - `test_unit_services.py` : **6 / 6 Passed (100%)**
  - `test_chain_of_responsibility.py` : **5 / 5 Passed (100%)**
  - `test_clearance_api.py` : **6 / 6 Passed (100%)**
  - `test_security_rbac.py` : **6 / 6 Passed (100%)**
  - `test_system_e2e_lifecycle.py` : **1 / 1 Passed (100%)**
  - **TOTAL: 73 / 73 TESTS PASSED (100% SUCCESS RATE)** in **2.16 SECONDS**.
- **Flake8 Static Code Quality: 0 Violations (100% Clean PEP 8)**.
- **Quality Status: PASSED ALL QUALITY GATES. APPROVED FOR OPERATIONAL DEPLOYMENT.**